KIYOON YOO

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INTRODUCTION

Applied AI researcher specializing in language model applications and agentic systems that enhance user experience through interactive AI. Currently developing of **embodied companion agents for PUBG (15M+ MAUs)**. Previously contributed to **Character Chat, a RAG-based chatbot used by 6M+ users** at Naver Webtoon. During my Ph.D, I published multiple times at top-tier venues including ACL, NAACL, and EMNLP with two oral presentations.

RESEARCH AND PROJECTS BY TOPICS

- LM applications: Real-time embodied agent [Krafton], Conversational AI [Naver Webtoon], Multimodal medical diagnosis [W2]
- Safety in Language Models: Watermarking texts and language model outputs [C2, C4], Adversarial attack and defense, backdoor attacks [C6, C7]
- Miscellaneous: Model compression [C3,C8], Representation learning [C5, W3]

WORK EXPERIENCE

Krafton AI (Applied Research Scientist) Sept. 2024 - Present

Developing PUBG ALLY, a real-time embodied companion agent that can play games alongside users; Early demo demonstrated in [CES 2025], and covered in [The Verge]

- **Agent Workflow:** Designed an embodied agent capable of fast, real-time gameplay and a dialogue/planning module handling user voice interaction over in-game mics.
- **Data Flywheel:** Built data collection loop with human-feedback and synthetic data augmentation pipelines enabling continuous model updates driven by user behavior.
- **Training** via imitation learning and model distillation to obtain real-time, small language model on consumer GPUs by leveraging quantization, achieving LLM-comparable performance.

Naver Webtoon (AI Engineer Intern) Aug. 2023 - Dec. 2023

Contributed to Character Chat, a production-scale AI chatbot mimicking webtoon character personalities and speech styles, used by over 6M users [media release]

- **Evaluation pipeline:** Led prompt engineering for LLM-as-evaluator workflows to systematically measure and reduce out-of-character responses, improving dialogue alignment and user immersion.
- **Pretraining pipeline:** Designed and implemented data collection, preprocessing, and pretraining pipeline for an in-house small Korean language model.

Conducted research in copyright protection

- LLM watermarking and copyright protection for text-to-image models (1 paper at NAACL 2024 and 1 at CVPR 2025). Placed 3rd in "N Innovation 2023", an internal competition for novel AI-driven services across Naver.

EDUCATION

Ph.D. in Intelligence and Information,
Department of Intelligence and Information, Seoul National University — Sept. 2019 - Aug. 2024
Supervised by Prof. Nojun Kwak

BS in Food Science and Biotechnology, (Minor: Industrial Engineering)
Department of Food and Animal Biotechnology, Seoul National University — March 2015 - Aug. 2019

PUBLICATIONS

<International Conference>

C1 Namhyuk Ahn, **KiYoon Yoo**, Wonhyuk Ahn, Daesik Kim, Seung-Hun Nam. "'Nearly Zero-Cost Protection Against Mimicry by Personalized Diffusion Models." CVPR, 2025.

C2 **KiYoon Yoo**, Wonhyuk Ahn, and Nojun Kwak. "Advancing beyond identification: Multi-bit watermark for large language models.", NAACL 2024. (long paper, oral presentation)

C3 Jangho Kim, Jayeon Yoo, Yeji Song, **KiYoon Yoo**, Nojun Kwak,"Dynamic Collective Intelligence Learning: Finding Efficient Sparse Model via Refined Gradients for Pruned Weights", ACM MM 2023.

C4 **KiYoon Yoo**, WonHyuk Ahn, Jiho Jang, Nojun Kwak. "Who Leaked My Document? Robust Natural Language Watermarking through Invariant Features", ACL 2023. (long paper)

C5 Jiho Jang, **KiYoon Yoo**, Seonhoon Kim, Kong Chaerin, Jangho Kim, Nojun Kwak, "Self-Distilled Self-Supervised Representation Learning", WACV 2023.

C6 **KiYoon Yoo**, Nojun Kwak, "Backdoor Attacks in Federated Learning by Rare Embeddings and Gradient Ensembling", EMNLP 2022. (long paper, oral presentation)

C7 **KiYoon Yoo**, Jangho Kim, Jiho Jang, Nojun Kwak, "Detection of Word Adversarial Examples in Text Classification: Benchmark and Baseline via Robust Density Estimation", Findings of ACL 2022. (long paper)

C8 Jangho Kim, **KiYoon Yoo**, Nojun Kwak, "Position-based Scaled Gradient for Model Quantization and Sparse Training", NeurIPS 2020.

<International Journals>

J1 NamHyuk Ahn, WonHyuk Ahn, **KiYoon Yoo**, DaeSik Kim, SeungHoon Nam. "Imperceptible Protection against Style Imitation from Diffusion Models.", IEEE Transactions on Multimedia, 2026. (Accepted)

J2 Hojun Lee, Donghwan Yun, Jayeon Yoo, **KiYoon Yoo**, et al., "Deep Learning Model for Real-Time Prediction of Intradialytic Hypotension", Clinical Journal of the American Society of Nephrology, 2021.

<Workshops>

W1 **KiYoon Yoo**, Wonhyuk Ahn, Yeji Song, and Nojun Kwak. "Exploring Causal Mechanisms for Machine Text Detection Methods.", Workshop on Trustworthy Natural Language Processing@NAACL 2024.

W2 Hyeonjin Kim, Min Kyu Kim, Jae Won Jang, **KiYoon Yoo**, Nojun Kwak, 'TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Dermatological Diagnosis", *2nd Place (English) for Multilingual & Multimodal Medical Answer Generation*, Workshop on ClinicalNLP@NAACL 2024.

W3 Inseop Chung, **KiYoon Yoo**, Nojun Kwak. "Open Domain Generalization with a Single Network by Regularization Exploiting Pre-trained Features.", Workshop on Data-centric Machine Learning Research@ICLR 2024.

W4 **KiYoon Yoo**, Nojun Kwak, "Backdoor Attacks in Federated Learning by Rare Embeddings and Gradient Ensembling", Workshop on Federated Learning for Natural Language Processing@ACL 2022.

ACADEMIC SERVICE

Reviewer

- ARR, EMNLP, ACL, ECCV, CVPR
- Neurocomputing

Program Committee

- Trustworthy Natural Language Processing@ACL2023
- AAAI 2025

AWARDS & SCHOLARSHIPS

Industrial Scholarship from Samsung, SAIT

- Sept. 2023 - Sept. 2024

Multilingual & Multimodal Medical Answer Generation Shared Task

- Workshop on ClinicalNLP@NAACL 2024
- 2nd place award in English track

N Innovation 2023 (3rd place in research track)

- Competition for innovative research across Naver